



Multionics profiling identifies molecular signatures of meat quality in sika deer cuts



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Sika deer (*Cervus nippon hortulorum*) meat is increasingly recognized for its nutritional and culinary value, yet the molecular mechanisms underlying its quality traits remain poorly understood. In this study, we performed integrated transcriptomic, proteomic, and metabolomic profiling of six representative muscle cuts from farmed sika deer ($n = 12$). Intercostal and trapezius muscles exhibiting enriched features related to tenderness, marbling, and flavor. Weighted gene co-expression network analysis identified two convergent multi-omics regulatory axes, in which transcriptional programs represented by ME6_trans and ME7_trans were linked to protein modules involved in muscle structural organization and contractile metabolism (ME3_protein), ultimately reflected in metabolite clusters associated with lipid utilization and energy homeostasis (ME2_metab). These coordinated transcription-to-protein-to-metabolite relationships suggest integrated regulation of extracellular matrix remodeling, muscle fiber functional properties, and lipid metabolic adaptation underlying region-dependent molecular variation in sika deer muscle. In addition, RT-qPCR and Western blot validation of selected key factors demonstrated expression patterns consistent with the multi-omics analyses, supporting the robustness of the identified molecular features. This study represents the first integrative multi-omics characterization of molecular variation across major muscle cuts in farmed sika deer, providing preliminary systems-level insights into coordinated metabolic and structural features associated with meat quality-related traits.

Sika deer (*Cervus nippon hortulorum*) is increasingly recognized as a premium meat source due to its high protein content, low fat, and favorable amino acid and fatty acid composition^{1,2}. China, as the world's largest producer of farmed sika deer with an estimated annual population of 700,000, has traditionally prioritized the harvesting of antlers for medicinal³. In recent years, the growing consumer demand for venison has prompted a strategic shift in the sika deer industry, transitioning from traditional antler-oriented practices toward meat-focused production systems⁴. Concurrently, sika deer meat has entered high-end food markets globally. In addition, regions such as New Zealand, Europe, and the United States have established integrated systems encompassing deer farming, meat quality evaluation, and product marketing⁵. These developments underscore the urgent need for systematic research on venison to support industry advancement and the comprehensive elucidation of meat quality characteristics.

However, research on deer meat quality remains sparse, particularly at the molecular level. To date, most studies have focused on macroscopic or physicochemical parameters, with limited application of high-throughput omics approaches. For example, a recent study by Jin et al. provided a comprehensive analysis of the physicochemical and nutritional properties of

meat from twelve sika deer cut parts, highlighting site-specific differences in pH, tenderness, amino acid content, and fatty acid composition⁴. However, such studies lack insight into the molecular regulatory mechanisms driving these differences. In recent years, high-throughput omics approaches—including transcriptomics, proteomics, and metabolomics—have been widely employed to elucidate meat quality traits in major livestock species such as cattle, pigs, and poultry^{6–8}. For instance, a 2024 study on beef cattle undergoing compensatory growth employed RNA-Seq to identify 57 differentially expressed genes and 31 enriched pathways related to pH, cooking loss, and fat content, highlighting the complement–coagulation–fibrinolysis system as a key regulatory network⁹. In pigs, a proteomic analysis used data-independent acquisition mass spectrometry to investigate the effects of chilling rate on pork quality, pinpointing several proteins involved in cut structure and energy metabolism¹⁰. Additionally, a metabolomic study of Jingyuan chickens identified 4643 differential metabolites across age groups, revealing links between oxidative phosphorylation, β -alanine metabolism, and cut lipid deposition with changes in pH_{45min} and color traits¹¹. Despite these advances in other livestock, comparable single-omics studies on sika deer are still absent, and integrated multi-omics investigations in deer meat have not yet been conducted. This gap underscores the need for a

comprehensive analysis of sika deer to elucidate the molecular basis of cut-specific meat quality traits, which would provide valuable insights for meat science and breeding strategies in cervids.

In response to this research gap, we employed an integrative multi-omics approach to analyze six representative cuts from farmed sika deer, incorporating transcriptomic, proteomic, and metabolomic data. By comparing samples from deer of different sexes and slaughter seasons, we further investigated how intrinsic (sex-related) and extrinsic (environmental) factors influence meat quality traits. After identifying differentially expressed genes, proteins, and differentially abundant metabolites, we performed functional enrichment analysis and constructed co-expression networks for each omics dataset using Weighted Gene Co-expression Network Analysis (WGCNA)¹², followed by cross-omics integration. This integrative analysis facilitated the identification of putative co-expression modules associated with regional variation in molecular features related to meat quality. Our study not only characterizes the molecular basis of cut-specific meat quality but also provides theoretical and data-driven support for future research on sika deer improvement and molecular breeding strategies.

Results

Overview of multi-omics analysis design and dimensionality reduction

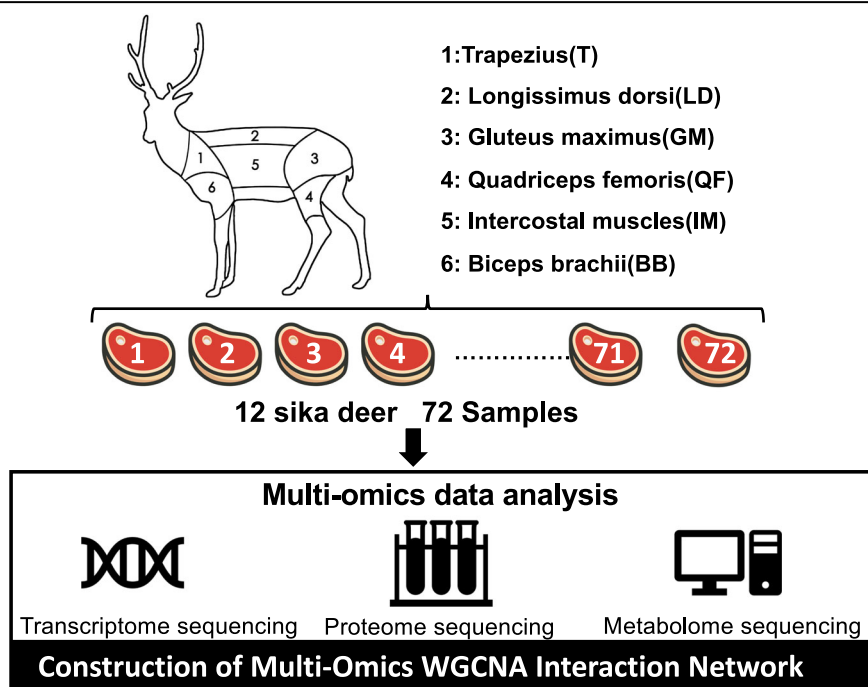
Yinshi Sun et al. revealed that LD and QF exhibit higher shear force and muscle fiber density⁴, whereas IM and T are characterized by greater intramuscular fat content and tenderness. BB and GM display moderate levels of water retention and collagen content, reflecting distinct and diverse meat quality traits among muscle regions. To explore the molecular mechanisms underlying meat quality differences across anatomical locations, we performed integrated transcriptomic, proteomic, and metabolomic profiling on six representative sika deer cuts, and the experimental design and workflow are shown in Fig. 1. After normalization, each omics level underwent dimensionality reduction to visualize overall expression variation (Supplementary Fig. 1A–C). Notably, no clear clustering by cut type was observed in transcriptomics (PCA), proteomics (PCA), or metabolomics (PLS-DA), indicating that subtle yet biologically meaningful differences may be embedded within specific regulatory features explored in subsequent analyses.

Transcriptional characteristics across cuts

To explore transcriptional differences among cut regions, we performed differential gene expression analysis using a “one-vs-rest” comparison strategy while adjusting for sex, sampling season, and individual animal identity. We identified a variable number of significantly differentially expressed genes (DEGs) across six cuts: BB (DEGs = 2), IM (DEGs = 26), QF (DEGs = 1), T (DEGs = 23), GM (DEGs = 4), and LD (DEGs = 5) (Fig. 2A and Supplementary Data 2). Subsequent KEGG enrichment analysis revealed that some DEGs were significantly associated with pathways implicated in meat quality traits (Fig. 2B and Supplementary Data 2). For example, the Protein digestion and absorption pathway includes several genes involved in cellular structure and cytoskeletal organization, such as collagens and actins, which may affect muscle fiber integrity and inter-muscular tenderness¹³. Additionally, the TGF- β signaling pathway, which is involved in muscle growth and differentiation, may play a role in regulating muscle development and tenderness¹⁴, and the PPAR signaling pathway, linked to lipid metabolism, could influence intramuscular fat deposition and flavor formation¹⁵. In addition, the ECM–receptor interaction pathway, involving collagens and integrins, may affect tissue stability and shear force, thereby contributing to meat toughness¹⁶. The IM region exhibited the greatest number of DEGs, including several upregulated genes closely linked to meat quality (Fig. 2C). *ADIPOQ* and *PLIN1*, key regulators of lipid metabolism, were significantly upregulated and may enhance marbling and juiciness^{17,18}. Meanwhile, upregulation of ECM-related genes such as *COL11A2*, *COL12A1*, *COL22A1*, *COMP*, *FMOD*, and *THBS4* suggests enhanced extracellular matrix stability, potentially improving cut structure and tenderness¹⁹. These results are consistent with Yinshi Sun et al.'s meat quality studies and highlight the functional relevance of these transcriptional differences in shaping region-specific meat characteristics⁴.

Finally, we conducted differential transcriptomic expression analyses across the six cuts regions in terms of sex and seasonal factors. In terms of sex, male samples exhibited more upregulated genes overall, with the GM cuts showing the most significant difference, containing 288 differentially expressed genes (Fig. 2D and Supplementary Data 2). KEGG pathway analysis revealed that male-upregulated genes were predominantly enriched in pathways related to protein processing in the endoplasmic reticulum and lipid metabolism (Fig. 2E and Supplementary Data 2), suggesting enhanced activity in protein synthesis and lipid regulation. Regarding seasonal effects,

Fig. 1 | Experimental design and multi-omics workflow. Overview of the experimental design. A total of 12 farmed sika deer were sampled, and six representative cuts regions were collected from each animal. Transcriptomic (RNA-Seq), proteomic (DIA-MS), and metabolomic (LC-MS/MS) analyses were performed on all cut samples to investigate molecular mechanisms underlying meat quality differences.



a total of 921 significantly different genes were identified, with 533 upregulated in spring and 388 in winter (Fig. 2F and Supplementary Data 2). KEGG enrichment indicated that spring-upregulated genes were mainly associated with protein synthesis and lipid-related signaling pathways, while winter-upregulated genes were enriched in carbon metabolism and circadian rhythm pathways (Fig. 2G and Supplementary Data 2). In summary, the differences in gene expression across cuts regions, sexes, and seasons reflect transcriptomic-level regulation of metabolic activity, structural integrity, and physiological functions in cuts tissues, ultimately contributing to the molecular basis underlying sika deer meat quality traits.

Proteomic profiles and functional insights

A total of 5367 proteins were detected (Supplementary Fig. 2A), with approximately 98% (5271) shared across all cuts (Fig. 3A), indicating a high overlap in protein types among different tissues. Based on this, differential expression analysis was conducted by comparing each cuts against the others, revealing differentially expressed proteins (DEPs) in each: BB (DEPs = 40), GM (DEPs = 56), IM (DEPs = 49), LD (DEPs = 60), QF (DEPs = 128), and T (DEPs = 63) (Fig. 4B and Supplementary Data 4). Functional enrichment showed region-specific proteomic patterns associated with cuts quality (Fig. 3C and Supplementary Data 3). LD-specific proteins were enriched in transcriptional regulation and arginine/proline metabolism pathways, indicating active protein synthesis and turnover, which may support tenderness and functional integrity²⁰. In contrast, downregulated proteins in LD were associated with lipid metabolism and the PI3K-Akt signaling pathway, suggesting a lower capacity for fat deposition and stress adaptation²¹. QF was enriched in glycolysis and pyruvate metabolism, reflecting high-energy metabolism that may influence cuts pH and flavor²². IM and GM were associated with lysosomal degradation and glycosaminoglycan metabolism, highlighting their potential role in maintaining extracellular matrix integrity and improving textural properties^{23,24}. BB and T showed upregulation in immune-related and developmental pathways, possibly reflecting stress response or repair mechanisms with less direct contribution to meat quality.

Regarding sex differences (Supplementary Fig. 2B and Supplementary Data 3), male samples exhibited upregulation in cAMP signaling and

morphine addiction pathways, suggesting more dynamic responses in neuromuscular regulation²⁵ (Fig. 3D and Supplementary Data 3). Female samples were enriched in ribosome biogenesis, basal transcription factors, and ABC transporters, indicating enhanced capacity for protein synthesis and metabolic stability²⁶. In terms of seasonal differences (Supplementary Fig. 2C and Supplementary Data 3), spring samples showed enrichment in transcriptional regulation and virus-related pathways, suggesting higher metabolic and renewal activity. In contrast, winter samples were enriched in carbon fixation, galactose metabolism, and starch/sucrose metabolism pathways, implying greater capacity for energy storage and sugar accumulation, which may enhance flavor and nutritional quality (Fig. 3E and Supplementary Data 3). Collectively, proteomic profiles across cuts types, sexes, and seasons highlight functional variations in metabolic regulation, structural maintenance, and lipid homeostasis, laying the molecular foundation for distinct meat quality traits.

Metabolomic differences and functional pathways

A total of 1124 metabolites were identified (Fig. 4A), primarily encompassing lipids and lipid-like molecules ($N = 278$), and fatty acyls were the most abundant subclass within lipids, suggesting robust activity in lipid synthesis, membrane stability, and flavor precursor metabolism, potentially influencing flavor, fat deposition, and metabolic activity²⁷. Differential expression analysis, adjusted for sex, sampling season, and individual animal identity, revealed variable numbers of significantly altered metabolites across cuts (Fig. 4B and Supplementary Data 4), with T and IM cuts exhibiting the highest counts, indicating more active metabolic features. Functional insights revealed region-specific enrichment of key metabolites and lipid classes closely related to meat quality. IM exhibited high levels of L-lysine and ADP, participating in amino acid and nucleotide metabolism, alongside fatty acyl compounds such as tiglic acid, oleamide, and propionylcarnitine, likely enhancing flavor, lipid aroma, and juiciness²⁸ (Supplementary Fig. 3A). In LD, the enrichment of L-Anserine and Lipoamide (linked to antioxidant defense and lipid transport), along with LPE 16:1 (a glycerophospholipid), may contribute to membrane stability and meat tenderness (Supplementary Fig. 3B). T was enriched in nucleotide metabolites (Guanosine, Uridine) and structural lipids (PC O-18:0), suggesting

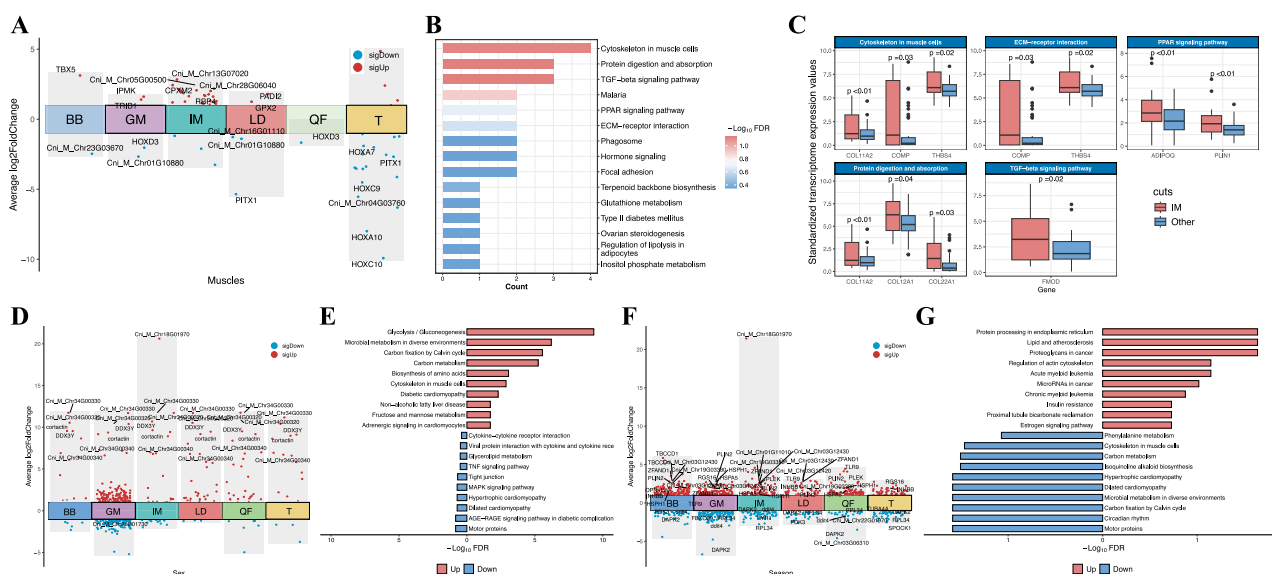


Fig. 2 | Transcriptomic characteristics across cut regions. **A** Differential expression analysis of six cuts regions. X-axis shows cuts types; red dots indicate upregulated genes, blue dots indicate downregulated genes. **B** KEGG pathway enrichment of all significantly differentially expressed genes across regions. X-axis shows gene counts; color represents adjusted p values. **C** Boxplots showing expression of genes in five key pathways for IM vs. other regions. Red indicates IM samples; blue indicates other cuts samples.

D Transcriptomic differences across cuts regions stratified by sex, same as above. **E** KEGG enrichment of DEGs upregulated in males (red) or females (blue) across all regions. X-axis shows $-\log_{10}$ adjusted p values. **F** Transcriptomic differences across cuts regions stratified by season, same as above. **G** KEGG enrichment of DEGs upregulated in spring (red) or winter (blue) across all regions. X-axis shows $-\log_{10}$ adjusted p values.

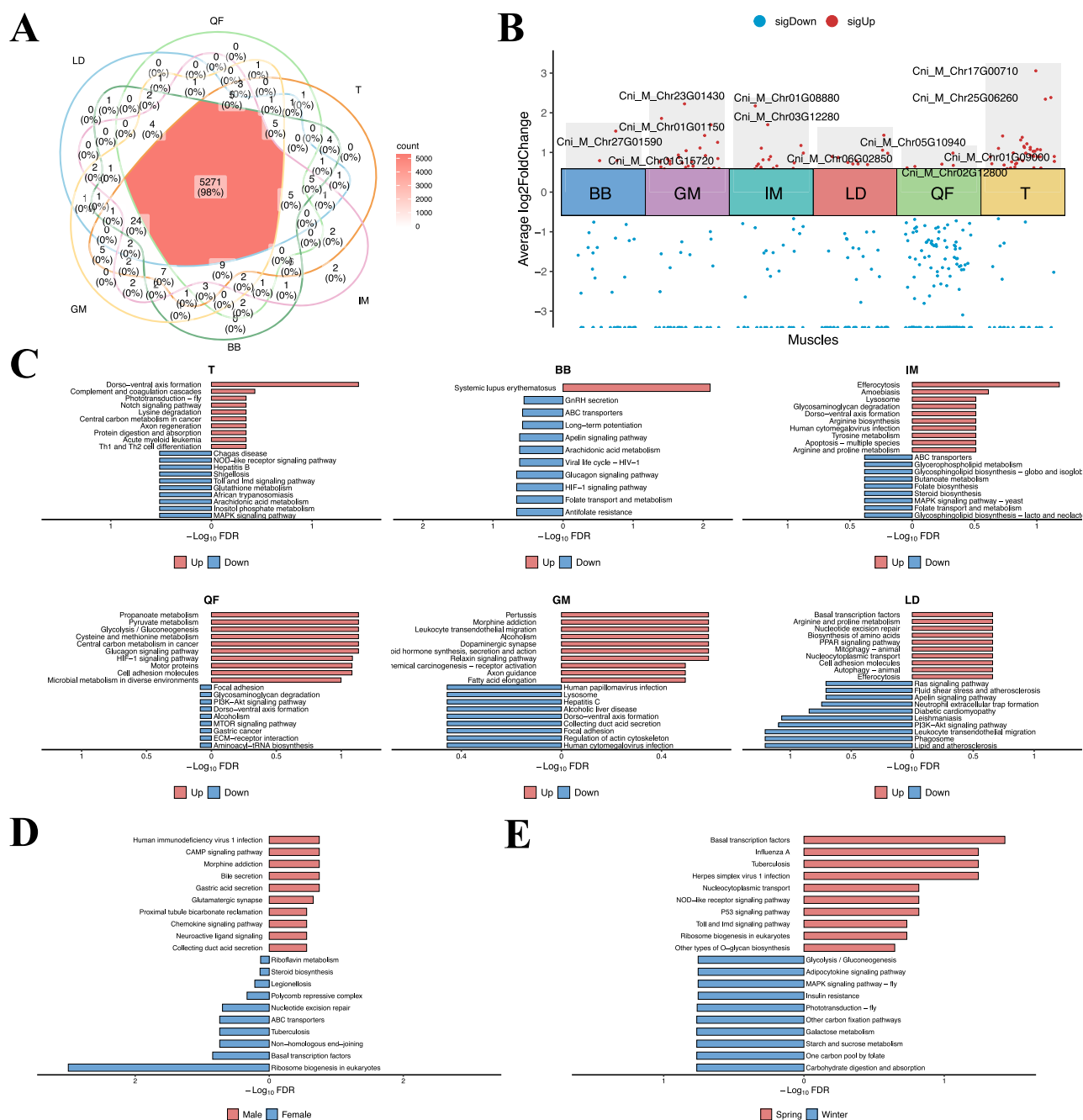


Fig. 3 | Proteomic profiles and functional insights across cuts regions in sika deer.

A Venn diagram illustrating the overlap of identified proteins among six cuts regions, with 98% shared across all samples. **B** Differential protein expression patterns for each region; X-axis shows cuts parts and Y-axis shows log2FC, with red and blue indicating significantly upregulated and downregulated proteins, respectively. **C** KEGG pathway enrichment of significantly altered proteins in each region; red

and blue bars represent up- and downregulated pathways, respectively. **D** KEGG enrichment of differentially expressed proteins between male and female samples; red and blue indicate pathways upregulated in male and female groups, respectively. **E** KEGG enrichment of differentially expressed proteins between spring and winter groups; red and blue indicate pathways upregulated in spring and winter samples, respectively.

enhanced energy balance and membrane integrity (Supplementary Fig. 2C). QF showed enrichment in Pyridoxal-5'-phosphate and diverse glycerophospholipids and fatty acyls (Supplementary Fig. 3D). Enrichment analyses of significantly altered metabolites revealed their involvement in nucleotide, lipid, amino acid, and energy metabolism (Fig. 4C and Supplementary Data 4), with lipid subclass analysis indicating dominant contributions from fatty acyls, glycerophospholipids, and sterol lipids (Fig. 4D and Supplementary Data 4), all of which are critical for tissue integrity, flavor, and meat texture.

Regarding biological variation, PLS-DA results demonstrated clear separations between sex and seasonal groups (Supplementary Fig. 3E, G). Seasonal analysis identified 294 differential metabolites (Supplementary Fig. 3F and Supplementary Data 4); KEGG pathways revealed that spring samples were enriched in nucleotide and general metabolic pathways, suggesting elevated biosynthetic and nutrient-processing activity, while winter samples showed enrichment in carbohydrate metabolism and nervous system pathways (Fig. 3E). Sex-based analysis identified 158 differential metabolites with region-specific patterns (Supplementary Fig. 3H and Supplementary Data 4), and GSEA results further showed male samples had

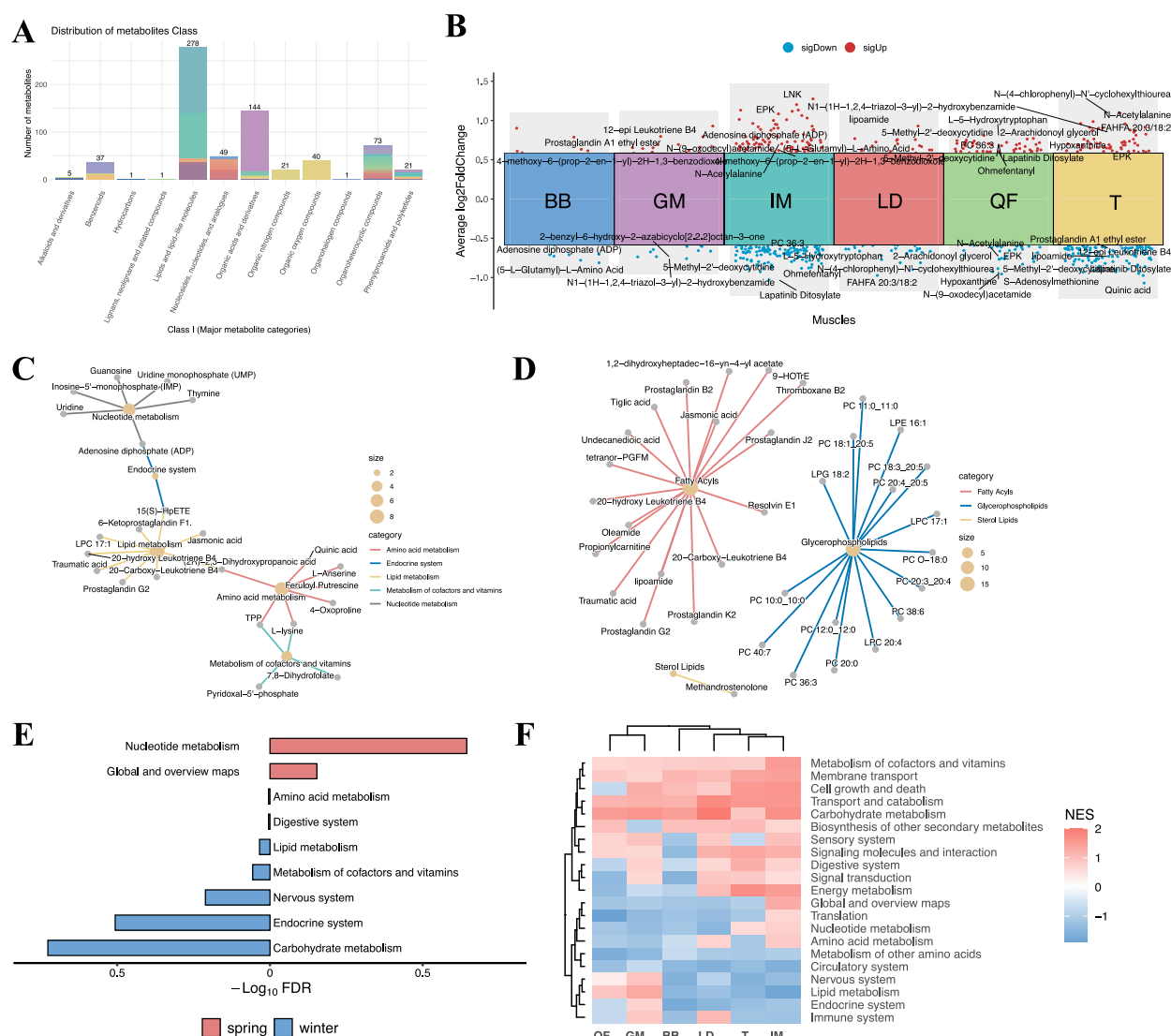


Fig. 4 | Metabolomic differences and enriched pathways across sika deer cuts regions. **A** Classification and subclass distribution of identified metabolites. **B** Differential metabolite expression across cuts regions; red indicates upregulation and blue indicates downregulation. **C** KEGG pathway enrichment network of significantly different metabolites. **D** LIPID MAPS enrichment network of significantly

different metabolites. **E** KEGG enrichment of metabolites upregulated in males (red) and females (blue); x-axis represents adjusted p values. **F** GSEA enrichment score heatmap across seasons; x-axis shows cuts regions, and color indicates NES based on FC.

enhanced activity in core metabolic pathways, while sex-biased differences in lipid and nervous system-related pathways varied across cuts (Fig. 4F). Collectively, these findings suggest that regional metabolic profiles and sex/season-dependent differences in structural lipids and core metabolic processes jointly contribute to the diversity of sika deer quality across different cuts.

WGCNA module-trait associations

Transcriptomic, proteomic, and metabolomic levels exhibit strong regulatory coordination, which shapes cuts structure, metabolic states, and flavor profiles. To investigate their interconnections, we performed WGCNA on each omics dataset using optimized soft thresholds (Supplementary Fig. 4A–C), identifying 25 transcriptomic (Fig. 5A), 14 proteomic modules (Fig. 5B), and 13 metabolomic (Fig. 5C). We then modeled module-cuts relationships using linear regression (adjusted for sex, sampling season, and individual animal identity). Results revealed multiple modules significantly associated with IM and T in the transcriptome (Fig. 5D), ME3, ME8, ME7,

ME10, and ME11 in the proteome (Fig. 5E), and ME2 and ME4 in the metabolome (Fig. 5F). Proteomic ME3, associated with glycolysis, amino acid biosynthesis, and cytoskeletal regulation, showed high expression in IM and low in QF, supporting roles in marbled, soft tissues²⁹ (Fig. 5G). ME8, enriched in structural and metabolic pathways including PPAR and adrenaline signaling, was highest in IM and lowest in QF, marking it as a key regulator of tenderness and lipid-rich meat quality³⁰ (Fig. 5H). The ME2 metabolic module, enriched in nucleotide and amino acid metabolism, was positively associated with QF and negatively with IM and T (Fig. 5I). This suggests that ME2 contributes to fast-twitch cuts properties such as higher metabolic activity, larger fiber size, and protein synthesis, while lower ME2 expression in IM and T aligns with slower oxidative fibers associated with tenderness and marbling³¹. ME4 was enriched in lipid metabolism, neuroendocrine signaling, and nervous system pathways. Its low expression in IM and T may promote fat accumulation and softness, while its elevated expression in QF supports firm cuts texture³² (Fig. 5J).

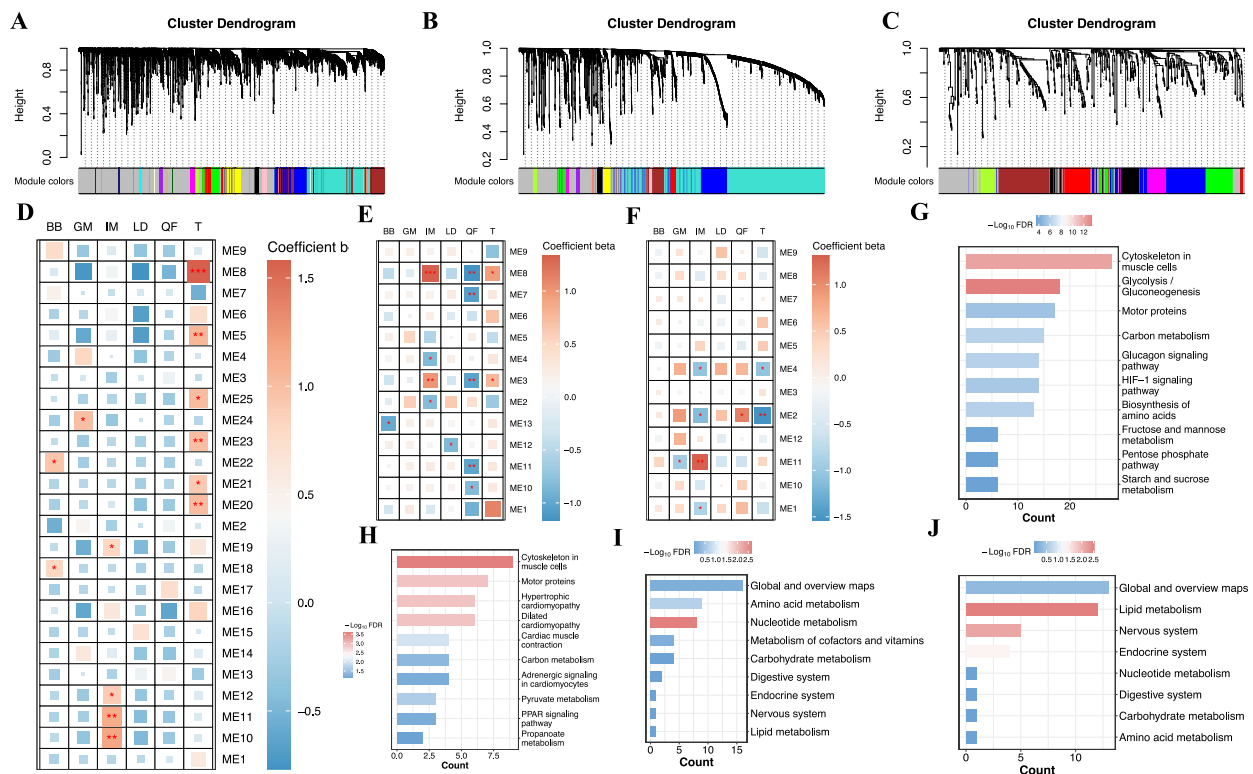


Fig. 5 | WGCNA across transcriptome, proteome, and metabolome. A–C Cluster dendrograms showing WGCNA modules identified from transcriptomic (25 module), proteomic (14 module), and metabolomic (13 module) data, respectively. D–F Correlation heatmaps between WGCNA modules and cuts regions for transcriptome, proteome, and metabolome. Red and blue indicate positive and negative correlations; *, **, and *** represent $p < 0.05$, 0.01, and 0.001, respectively. G KEGG

enrichment of proteins in ME3 proteomic module, same as above. H KEGG enrichment of proteins in ME8 proteomic module, same as above. I KEGG enrichment of metabolites in ME2 metabolic module. X-axis indicates the number of metabolites enriched in each pathway, the Y-axis shows KEGG pathway names, and the color gradient represents adjusted p values. J KEGG enrichment of metabolites in ME4 metabolic module, same as above.

In summary, multi-omics WGCNA identified functional modules linked to specific cuts, providing system-level insights into the molecular regulation of regional meat quality traits.

Multi-omics integration reveals key regulatory hubs

To explore multi-level regulatory mechanisms, we constructed a cross-omics interaction network based on WGCNA module correlations and identified several significantly associated module pairs across transcriptomic, proteomic, and metabolomic levels (Fig. 6A). Among them, ME4_metab and ME8_protein exhibited strong negative correlation and were both highly associated with the IM region, suggesting that this cuts area features lipid-driven and endocrine-regulated activity, while having lower energy and structural metabolism—traits consistent with soft texture and rich flavor. Moreover, ME3_protein and ME2_metab formed two integrated regulatory chains with ME6_trans and ME7_trans (ME6_trans–ME3_protein–ME2_metab and ME7_trans–ME3_protein–ME2_metab). ME6_trans was enriched in oxidative phosphorylation and glycolysis (Fig. 6B), while ME7_trans focused on carbon metabolism and lipid biosynthesis (Fig. 6C), indicating their roles in energy supply, cuts fiber formation, and fat deposition.

We further constructed a tri-omics regulatory network (Fig. 6D) and identified key hub molecules. Transcriptomic hub genes like *COQ9* and *NDUFA8* (energy metabolism) were highly expressed in LD, BB, and GM, whereas *ITGA6* and *WWC1* (linked to structural integrity and lipid metabolism) were enriched in IM and QF, reflecting their roles in fat accumulation and cuts quality formation (Fig. 6E). In the proteomic level, *TBB1*, *TNNC1*, *TNNT1*, and *MYL10* emerged as key hub proteins, showing high expression in IM and T—cuts regions known for higher fat content and tender texture—and lower levels in the firmer BB and QF regions. *TBB1*

contributes to cytoskeletal integrity, *TNNC1* and *TNNT1* regulate slow-twitch fiber contraction, and *MYL10* influences cuts flexibility and lipid metabolism. Their elevated expression may support the softness, elasticity, and marbling features of IM and T, reinforcing their superior tenderness and flavor (Fig. 6F). At the metabolite level, lipid oxidation products such as Prostaglandin G2 and ($\pm$)11-HETE were downregulated in IM and T, potentially contributing to tenderness and flavor. Conversely, 19(R)-hydroxy PGE2 was elevated in QF, likely supporting rapid metabolism and firmer texture (Fig. 6G).

In summary, our multi-omics WGCNA integration identified a set of key regulatory modules and hub molecules associated with cuts region-specific traits. In particular, the two core interaction chains—ME6_trans–ME3_protein–ME2_metab and ME7_trans–ME3_protein–ME2_metab—highlight the coordinated regulation of lipid metabolism, energy supply, and tissue structure, representing major findings of this study and offering valuable targets for meat quality improvement and molecular breeding.

RT-qPCR and Western blot validation of key factors

FMOD, THBS4, and PLIN1 were selected as representative IM-enriched genes, while *TNNC1* and *TNNT1* were chosen as hub proteins from the multi-omics network, and validation was performed using samples from adult male animals with three independent biological replicates for each muscle region. RT-qPCR analysis (Fig. 7A–E) demonstrated that *TNNC1* and *TNNT1* exhibited consistently higher expression in IM, while *FMOD*, *THBS4*, and *PLIN1* displayed region-dependent expression patterns with relatively elevated levels in IM compared to several other cuts. These trends were concordant with the transcriptomic result. Western blot analysis (Fig. 7F–H) further confirmed differential protein abundance of *FMOD*,

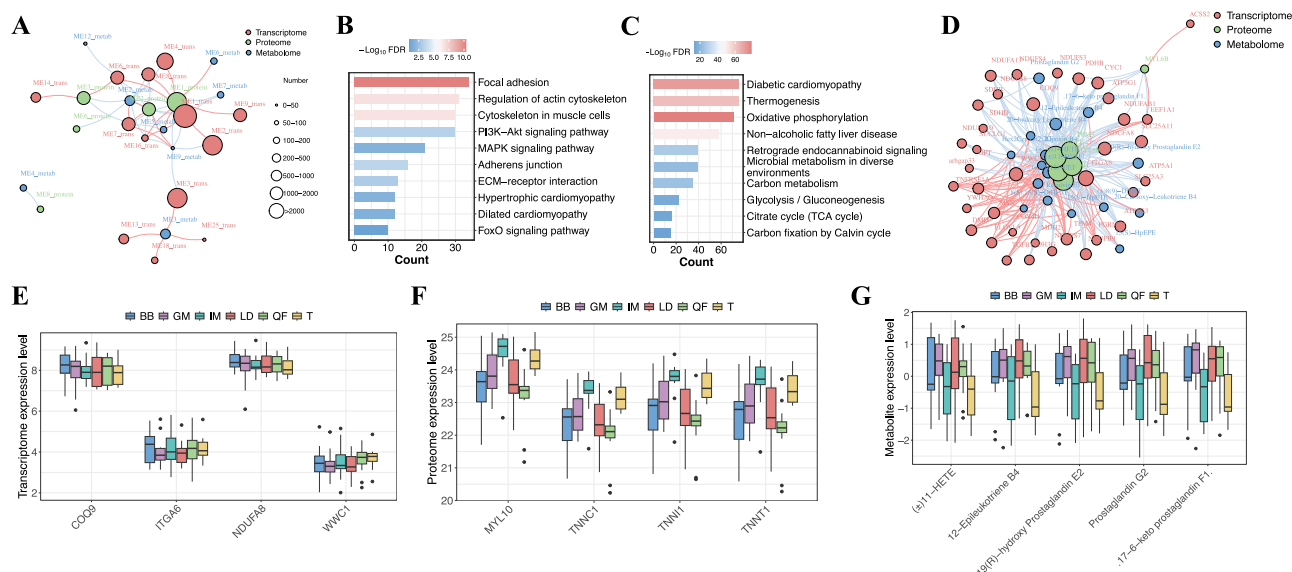


Fig. 6 | Multi-omics integration reveals key regulatory hubs associated with meat quality in different cuts regions. **A** Correlation network among WGCNA modules derived from transcriptomics (red nodes), proteomics (green nodes), and metabolomics (black nodes). Node size represents the number of features in each module; edge color indicates the correlation direction (red: positive; blue: negative), with all edges showing significant correlations ($|r| > 0.5$, $p < 0.05$). **B** KEGG pathway enrichment of transcriptomic module ME6. **C** KEGG pathway enrichment of transcriptomic module ME7. **D** Multi-omics correlation networks for two

regulatory cascades: ME6_trans–ME3_protein–ME2_metab and ME7_trans–ME3_protein–ME2_metab. Node color indicates data level (red: transcriptome, green: proteome, black: metabolome); node size reflects node connectivity (degree), and edge color indicates the correlation direction (red: positive; blue: negative) with $|r| > 0.3$ and $p < 0.05$. Expression boxplots of hub feature, include transcriptomics (**E**), proteins (**F**) and metabolites (**G**) from the two regulatory cascades across cuts regions. X-axis shows feature names, Y-axis represents normalized expression values, and color indicates cuts type.

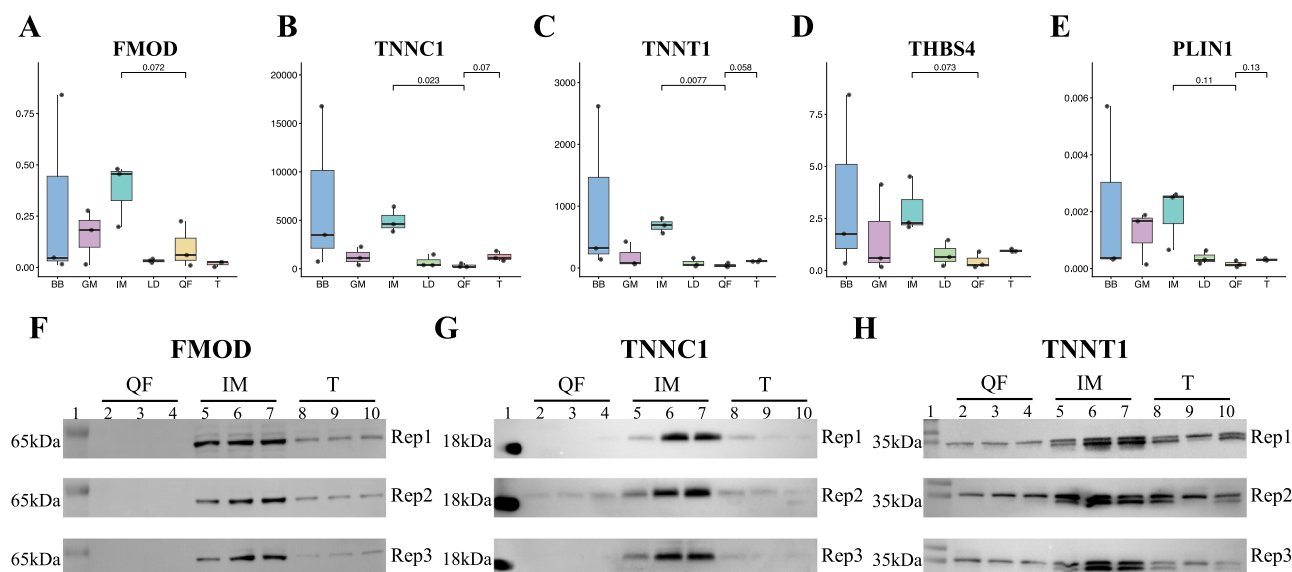


Fig. 7 | RT-qPCR and Western blot validation of key factors. RT-qPCR analysis of *FMOD* (**A**), *TNNC1* (**B**), *TNNT1* (**C**), *THBS4* (**D**), and *PLIN1* (**E**) across six cuts regions. The x-axis represents cuts regions, and the y-axis shows relative expression

levels. **A** Western blot analysis of *FMOD*, *TNNC1*, and *TNNT1* in QF, IM, and T, with three biological replicates per group.

TNNC1, and *TNNT1* across QF, IM, and T, with the highest expression observed in IM and relatively elevated levels in T, consistent with the proteomic profiles. Original blots are shown in Supplementary Figs. 5–7. Collectively, these cross-platform validations reinforce the reliability of the multi-omics findings and support the relevance of these molecules to cut-specific molecular characteristics associated with meat quality traits.

Discussion

Sika deer meat has attracted increasing attention because of its high protein content, low fat level, and favorable nutritional composition. Although previous research has reported differences in physicochemical traits among venison cuts, the molecular mechanisms associated with these differences are still not fully understood^{1–3}. In recent years, omics technologies have been increasingly applied in cattle, pigs, and poultry to investigate biological

processes associated with tenderness, fat deposition, and flavor development^{6–10}. Comparable studies in sika deer are still limited, especially in integrating multiple omics layers. In this study, transcriptomic, proteomic, and metabolomic datasets were analyzed to characterize molecular variation among six representative muscle cuts from farmed sika deer.

Different muscle cuts exhibited varying degrees of molecular variation across the transcriptomic, proteomic, and metabolomic datasets. At the transcriptomic level, IM showed relatively more differentially expressed genes, including *ADIPOQ* and *PLIN1*, which are associated with lipid metabolism and intramuscular fat deposition^{17,18}. Several extracellular matrix-related genes, including *COL11A2*, *COL12A1*, *COL22A1*, *COMP*, *FMOD*, and *THBS4*, were also relatively highly expressed in IM. These genes have been reported to participate in extracellular matrix organization and structural maintenance in skeletal muscle¹⁹. Pathways including PPAR signaling, ECM–receptor interaction, and TGF- β signaling were also enriched among different muscle regions. Previous studies have linked PPAR signaling to lipid metabolism and intramuscular fat deposition, whereas ECM-related pathways may contribute to tissue structure and tenderness-associated properties^{13,19}. At the proteomic level, QF was mainly enriched in glycolysis and pyruvate metabolism pathways, which may influence muscle energy metabolism and meat quality characteristics²⁷. In contrast, IM and GM were associated with lysosomal degradation and glycosaminoglycan metabolism, pathways potentially related to extracellular matrix maintenance and tissue properties^{23,24}. In addition, proteins such as *TNNC1*, *TNNT1*, and *MYL10* showed relatively high abundance in IM and T, consistent with their roles in muscle contraction and slow-twitch fiber characteristics. Metabolomic analysis identified fatty acyls, glycerophospholipids, and sterol lipids as major differential metabolite classes among muscle regions^{27,28}. IM contained relatively high levels of metabolites such as tiglic acid, oleamide, and propionylcarnitine, which may be associated with lipid metabolism and flavor-related characteristics²⁸.

Cross-omics integration identified several correlated transcriptomic, proteomic, and metabolomic modules associated with regional differences among muscle cuts. Among these, ME3_protein and ME8_protein were enriched in pathways including glycolysis, amino acid biosynthesis, cytoskeletal regulation, and PPAR signaling. Previous studies have linked these processes to tenderness-related traits and intramuscular fat deposition^{29,30}. Metabolomic modules ME2 and ME4 displayed opposite association patterns between QF and IM/T, suggesting differences in energy metabolism and lipid-associated processes among muscle regions^{31,32}. Two correlated module chains, ME6_trans–ME3_protein–ME2_metab and ME7_trans–ME3_protein–ME2_metab, connected transcriptomic, proteomic, and metabolomic features associated with muscle structure and metabolism. In addition, hub molecules including *COQ9*, *NDUFA8*, *ITGA6*, *TNNC1*, and *TNNT1* displayed region-specific expression patterns. Finally, RT–qPCR and Western blot analyses of genes associated with lipid deposition, extracellular matrix-related characteristics, and muscle contraction features (*FMOD*, *THBS4*, *PLIN1*, *TNNC1*, and *TNNT1*) further supported the expression patterns observed in the multi-omics analyses.

Several limitations should also be considered. First, this study mainly focused on molecular profiling and did not include direct, detailed histological analyses of meat quality traits. Second, the relationships identified by WGCNA were based on correlation analyses and cannot fully demonstrate causal regulation among genes, proteins, and metabolites. Although RT–qPCR and Western blot analyses supported several of the observed expression patterns, validation was limited to selected targets and representative muscle regions.

In conclusion, this study provides an integrative characterization of molecular differences across six major muscle cuts in farmed sika deer. Integrated transcriptomic, proteomic, and metabolomic analyses revealed region-dependent molecular differences among muscle regions, with intercostal and trapezius muscles showing coordinated molecular features related to lipid metabolism, extracellular matrix organization, muscle contraction, and energy utilization. Cross-omics integration further identified two correlated module chains, ME6_trans–ME3_protein–ME2_metab and

ME7_trans–ME3_protein–ME2_metab, linking transcriptional variation to structural and contractile protein modules and downstream metabolic characteristics. The region-dependent expression patterns of *FMOD*, *THBS4*, *PLIN1*, *TNNC1*, and *TNNT1* were further supported at the transcript and protein levels by RT–qPCR and Western blot analyses, providing independent validation of selected multi-omics findings. These findings provide candidate molecular features associated with variation among sika deer muscle cuts and establish a reference framework for future studies integrating direct meat quality measurements and functional validation.

Methods

Sample collection

A total of 72 cuts were obtained from 12 healthy adult sika deer (6 males and 6 females), all raised under standardized feeding and management conditions at Shuangyang Deer Quality Breed Breeding Co., Ltd. (Changchun, China). All animals were selected with comparable developmental status and body condition and were maintained under standardized intensive housing conditions with routine health monitoring to minimize biological heterogeneity. The sample size was determined to ensure sufficient statistical power while adhering to the principles of animal welfare. No animals or samples were excluded from the analysis. Due to the seasonal reproductive characteristics of sika deer, estrus generally occurs in autumn, and parturition is concentrated mainly between May and June each year. All animals in the spring group were slaughtered on April 1, 2024, at 22–23 months of age, whereas all animals in the winter group were slaughtered on December 23, 2024, at 29–30 months of age. Therefore, when sika deer from the same birth cohort are slaughtered in different seasons, a difference of about 6 months in biological age is unavoidable. Within each group, all animals were slaughtered on the same day under uniform procedures to maintain consistency in their physiological condition, slaughter handling, and postmortem treatment. Prior to sample collection, all deer were transported to the Shilu Deer Industry Group Co., Ltd. in Shuangyang District, Changchun, Jilin Province, where slaughtering was performed under uniform protocols. Six distinct cuts—longissimus dorsi (LD), intercostal muscles (IM), trapezius (T), biceps brachii (BB), gluteus maximus (GM), and quadriceps femoris (QF)—were dissected postmortem. Detailed sample information is provided in Supplementary Data 1. According to the specific requirements of transcriptomic, proteomic, and metabolomic analyses, tissue samples were appropriately processed and preserved, including storage in RNA-later solution and/or immediate snap-freezing in liquid nitrogen. All procedures involving animals were reviewed and approved by the Institutional Animal Care and Use Committee of Northeast Forestry University, China (Approval No. 20220409). This study was reported in accordance with the ARRIVE guidelines (Animal Research: Reporting of In Vivo Experiments).

Transcriptomic sequencing and data analysis

Total RNA was extracted from cuts tissue samples collected from different anatomical regions of sika deer using the Trizol method and quantified with the NanoDrop® 2000 spectrophotometer (Thermo, USA). Genomic DNA was removed with RNase-free DNase I and the mRNA libraries were constructed using the TruSeq Stranded mRNA Library Prep Kit (Illumina) with 5 μ g of input RNA. Fragmentation, reverse transcription, adapter ligation, and amplification were conducted following the manufacturer's protocol. A 150 bp insert size was selected using AMPure XP beads (Beckman Coulter), and library quality was validated on the Agilent Bioanalyzer 2100 system. Sequencing was performed on the Illumina NovaSeq 6000 platform.

Raw reads were filtered using FASTP (v0.20.1)^{33,34} to remove adapters and low-quality sequences. Clean reads were aligned to the sika deer reference genome (GCA_040085125.1)¹² using HISAT2 (v2.0.4)³⁵. Gene-level counts were generated with HTSeq³⁶, and expression levels were calculated in FPKM (fragments per kilobase of transcript per million mapped reads) using the formula: $FPKM = (\text{read count} \times 10^9) / (\text{total mapped reads} \times \text{gene length})$.

Differential gene expression analysis was conducted using the DESeq2³⁷ (v1.42.0) based on read count data obtained from RNA-seq. To ensure statistical robustness and to avoid potential pseudoreplication arising from multiple muscle samples collected from the same animal, sex, sampling season, and individual animal identity were included as covariates in the design formula. For each cuts type, pairwise comparisons against the remaining groups were performed, and differentially expressed genes (DEGs) were defined using the threshold of adjusted *p* value (*padj*) < 0.05 and absolute log₂ fold-change (log₂FC) > 1. For dimensionality reduction, transcript per million (TPM) values were calculated by normalizing FPKM values across genes within each sample³⁸. The top 2000 most variable genes were selected for principal component analysis (PCA) to assess global expression patterns across samples.

Proteomic profiling and data analysis

Cuts tissue samples from different anatomical regions of sika deer were ground in liquid nitrogen and lysed using SDT buffer supplemented with DTT³⁹. After ultrasonication on ice and thermal denaturation at 95 °C, samples were centrifuged, and supernatants were alkylated with IAM in the dark. Proteins were precipitated with pre-cooled acetone and washed with cold acetone before being re-dissolved in dissolution buffer. Protein concentrations were measured using a Bradford assay, and protein integrity was assessed via SDS-PAGE. For digestion, proteins were denatured in 8 M urea with 100 mM TEAB (pH 8.5), and trypsin was added for a two-step digestion at 37 °C. After acidification, peptides were desalted using C18 columns and dried under vacuum. Dried peptides were reconstituted, centrifuged, and subjected to LC-MS/MS analysis using a Vanquish™ Neo UHPLC system coupled to a Thermo Orbitrap Astral mass spectrometer. Peptides were separated on a PepMap™ Neo C18 column with a linear gradient of acetonitrile containing 0.1% formic acid. DIA acquisition was performed with 300 variable windows (*m/z* 380–980) and resolution settings of 240,000 (MS1) and 80,000 (MS2).

Raw DIA data were analyzed using DIA-NN software (v2.2.0) in DirectDIA mode⁴⁰. A project-specific protein database was used for spectral matching, with iRT peptides for retention time calibration. Search parameters included 10 ppm precursor mass tolerance and 0.02 Da fragment tolerance. Cysteine residues were carbamidomethylated, and methionine oxidation and N-terminal modifications were considered. Up to two missed cleavages were allowed. Peptide-spectrum matches (PSMs) were filtered to retain only those with ≥99% confidence, and false discovery rate (FDR) thresholds were set at 1% for both peptides and proteins.

We performed principal component analysis to reduce dimensionality and visualize the distribution of proteomic expression profiles across different cuts regions. Protein abundance data were standardized using *z*-score transformation to ensure comparability across samples. Subsequently, two-sample *t*-tests were conducted to identify proteins with significant abundance differences between each cuts region and all others. FC was calculated as the ratio of mean expression levels between groups, providing a quantitative measure of differential expression.

Metabolomic profiling and data analysis

Approximately 100 mg of cuts tissue from different anatomical regions of sika deer was ground in liquid nitrogen and extracted with prechilled 80% methanol by vortexing. The homogenates were incubated on ice for 5 min and centrifuged at 15,000 × *g*, 4 °C for 20 min. The supernatant was diluted with LC-MS grade water to a final methanol concentration of 53%, then centrifuged again under the same conditions. The resulting supernatants were transferred to autosampler vials for LC-MS/MS analysis⁴¹. Chromatographic separation was performed on a Hypersil Gold C18 column (100 × 2.1 mm, 1.9 μm) using a Vanquish UHPLC system (Thermo Fisher Scientific), coupled with a Q Exactive™ HF or HF-X Orbitrap mass spectrometer (Thermo Fisher Scientific). A 12-min linear gradient of 0.1% formic acid in water (eluent A) and methanol (eluent B) was applied at a flow rate of 0.2 mL/min. Mass spectrometry was operated in both positive and negative ion modes, with a spray voltage of 3.5 kV, capillary

temperature of 320 °C, and sheath/auxiliary gas rates of 35 and 10 L/min, respectively.

Raw MS data were processed using Compound Discoverer 3.3 (Thermo Fisher) for peak detection, alignment, and quantification. Main parameters included a 5 ppm mass tolerance, 30% signal intensity tolerance, and normalization to total ion intensity. Metabolites were annotated by matching MS/MS spectra against mzCloud, mzVault, and MassList databases. Quality control filtering excluded compounds with a relative standard deviation >30% in QC samples. Identified metabolites were annotated using KEGG⁴², and LIPIDMaps⁴³ databases.

Metabolite profiling data were normalized and analyzed through Partial Least Squares Discriminant Analysis (PLS-DA) using the mixOmics (v6.26.0)⁴⁴ in R to identify cuts-specific metabolic signatures. The PLS-DA model was used to evaluate the contribution of each metabolite to the discrimination between cuts types. Metabolites with Variable Importance in Projection (VIP) scores >1 were selected as discriminant features. Differential abundance analysis was conducted with limma (v3.58.1)⁴⁵, applying thresholds of 1.5-FC and *p* value < 0.05, while controlling for sex, sampling season, and individual animal identity as covariates.

KEGG pathway enrichment analysis

KEGG pathway enrichment analysis was conducted using KOBAS (v2.0)⁴⁵, where all differentially expressed genes were first annotated to KEGG Orthology (KO)⁴² terms using *Bos taurus* (bta) as the reference species. Subsequently, enrichment testing via hypergeometric distribution using the R package clusterProfiler (v4.19.4) to identify significantly overrepresented biological pathways (adjusted *p* value < 0.05).

WGCNA-based module identification

WGCNA was applied to transcriptomic, proteomic, and metabolomic datasets to identify co-expressed modules and explore their associations with cuts regions. WGCNA package (v1.72.5) was applied, we first pre-processed the data and removed outlier samples. The optimal soft-thresholding power was selected using pickSoft-Threshold to construct a scale-free network. An adjacency matrix was generated and transformed into a topological overlap matrix (TOM), followed by hierarchical clustering and dynamic tree cut to identify discrete modules. Each module was summarized by its module eigengene. We then assessed the correlation between module eigengenes and cuts parts, identifying trait-related modules, which were further analyzed for KEGG pathway enrichment. Finally, we calculated pairwise correlations between module eigengenes across the three omics levels. Module pairs with $|r| > 0.5$ and *p* < 0.05 were selected to construct a multi-omics regulatory network, highlighting cross-omics functional links related to cuts characteristics.

RT-qPCR

RT-qPCR was used to quantify the expression of key genes (FMOD, TNNC1, TNNT1, THBS4, and PLIN1) across six muscle regions, with three biological replicates per group. Total RNA was extracted using TRIzol reagent (Takara, Japan), and reverse transcription was conducted using the PrimeScript™ FAST RT reagent Kit with gDNA Eraser (Takara, Japan) according to the manufacturer's instructions. Quantitative PCR was carried out using TB Green® Premix Ex Taq™ II (Tli RNaseH Plus) on a SLAN-96S Real-Time PCR system (Hongshi Medical Technology, China). Each sample was analyzed with three technical replicates. RPLP0 was used as the internal reference gene for normalization. Relative gene expression levels were calculated using the $2^{-\Delta\Delta Ct}$ method after quality control, where ΔCt represents the difference between target and reference genes, and $\Delta\Delta Ct$ is defined as the difference between experimental and control groups. Primer sequences were designed using Primer3 and are listed in Supplementary Data 5.

Western blotting

Western blotting was performed to evaluate the protein abundance of FMOD, TNNC1, and TNNT1 in representative muscle tissues, including

QF, IM, and T. Detailed information on antibodies is provided in Supplementary Data 6. Samples were rinsed with cold PBS and mechanically homogenized. Total proteins were extracted using RIPA buffer supplemented with protease inhibitors, followed by incubation on ice with intermittent mixing to ensure complete lysis. The lysates were then centrifuged, and the supernatants were collected for further analysis. Equal amounts of protein were separated by SDS-PAGE and transferred onto PVDF membranes. After blocking to minimize non-specific binding, membranes were incubated with primary antibodies against FMOD, TNNC1, and TNNT1 at 4 °C, followed by incubation with HRP-conjugated secondary antibodies at room temperature. Protein signals were detected using a chemiluminescence imaging system. Band intensities were quantified using image analysis software, and relative protein levels were calculated by normalization to the internal reference protein. All experiments were performed with three independent biological replicates to ensure reproducibility and reliability of the results.

Statistical analysis and data visualization

All statistical analyses were conducted using R software (v4.3.1). Pearson correlation was applied to evaluate inter-module relationships across omics levels, while Spearman correlation was used to assess associations between WGCNA modules and cuts regions, with $p < 0.05$ considered significant. Metabolomic pathway activity was assessed using the GSEA function from the R package clusterProfiler (v4.19.4), based on log2FC values across cuts groups. All visualizations were generated using R packages, including ggplot2 (v3.5.1)⁴⁶, ComplexHeatmap (v2.18.0)¹⁴, patchwork (v1.3.0.9)¹⁶, and scRNAtoolVis (v0.1.0).

Data availability

The datasets generated and analyzed during this study have been deposited in the following public repositories and are available upon request from the corresponding author: Metabolomics data have been deposited in the MetaboLights database (<https://www.ebi.ac.uk/metabolights/>) under the accession number MTBLS12900. Proteomics data are available via the iProX database (<https://www.iprox.cn/>) with the ProteomeXchange accession number PXD067526. Transcriptomics data have been deposited in the NCBI Sequence Read Archive under the accession number PRJNA1213237.

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Author contributions

Heping Li and Lan Tang designed the study and conducted the literature review; Lan Tang, Derui Zhao, Yukai Ma, and Tao Zhu collected the samples; Lan Tang and Qianghui Wang performed the data analysis; Lan Tang, Qing Wang, and Haihua Xing wrote the manuscript; Lan Tang and Yukai Ma edited; Heping Li, Lan Tang, Qing Wang, and Haihua Xing revised the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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